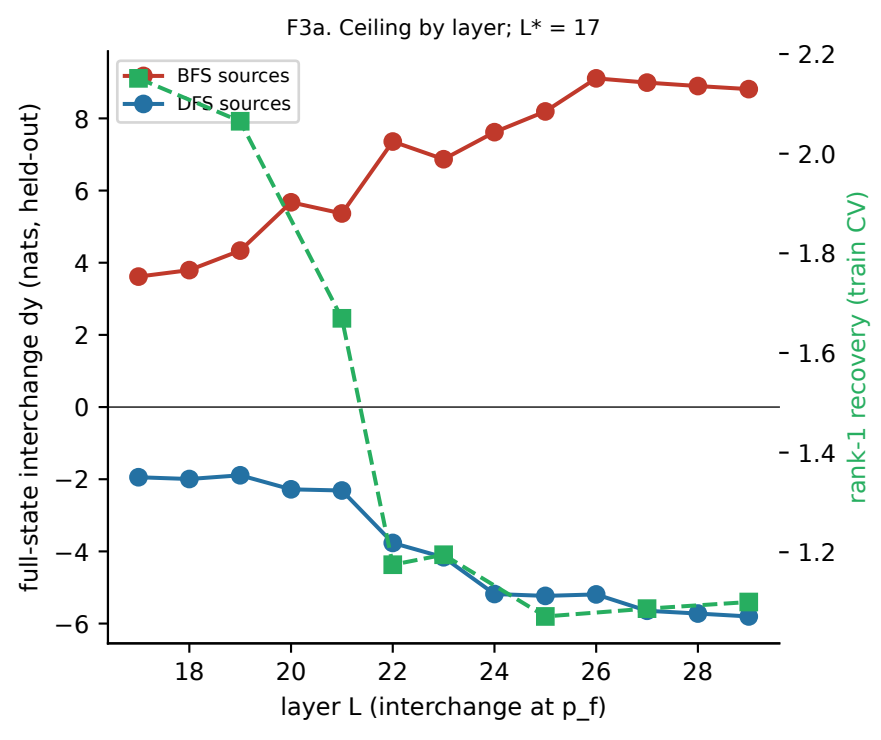
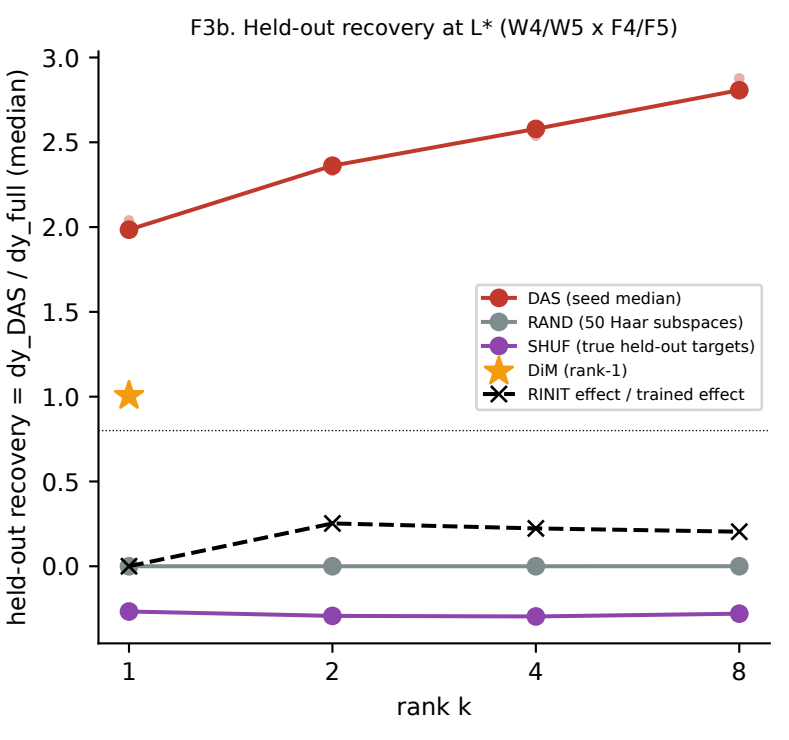

F3a. Ceiling by layer; L* = 17
BFS sources
DFS sources
full-state interchange dy (nats, held-out)
rank-1 recovery (train CV)
layer L (interchange at p_f)
8
6
4
2
0
−2
−4
−6
2.2
2.0
1.8
1.6
1.4
1.2
18
20
22
24
26
28


F3b. Held-out recovery at L* (W4/W5 x F4/F5)
held-out recovery = dy_DAS / dy_full (median)
rank k
DAS (seed median)
RAND (50 Haar subspaces)
SHUF (true held-out targets)
DiM (rank-1)
RINIT effect / trained effect
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1
2
4
8